SAY WATT ROBOTICS · FIELD LOG

Week 2.2 — The List Before the Parts — A Bill of Materials, and a Sizing Pass That Said Go Smaller

Sunday, September 27, 2026 · 3:00 – 5:00 PM · 11 students

Eleven of us at 2.2 — Trenton, Rhylan, Oliver, Eli, Annabelle, Liam, Maddox, Rebecca, Tyler, Evan, and Adrian — from 3:00 to 5:00. This was the session where the profiler stopped being a rubric and started being a list. The team built the float's first bill of materials. It isn't finished — a few things are still missing — but the bulk of it is on the page, and a bill of materials with gaps you can see is worth more than a design carried around in someone's head. The buoyancy calculations started too, and the first pass did exactly what a first pass is for. Sized around a six-inch PVC pipe, the math said the float would have to displace far more water than the team could ever ballast to bring it to neutral. So the answer came back: go smaller, to an outside diameter of roughly four and a half inches. Nobody had to buy or cut a single piece of pipe to learn that. The payload track kept developing the manipulator design alongside it.

11	1st	6 in	4.5 in
STUDENTS AT THE SESSION	BILL OF MATERIALS FOR THE PROFILER	PIPE IN THE FIRST SIZING PASS — TOO BIG	OUTSIDE DIAMETER THE MATH POINTED TO INSTEAD

WHAT WE LEARNED

- A bill of materials is a design document. Writing every part down is how you find the parts you forgot — and how the CFO learns what the float costs before the receipts arrive.
- Buoyancy is a volume problem. A float is neutral when it weighs exactly as much as the water it pushes aside, so every extra bit of diameter is water you have to match with weight.
- A first pass is supposed to be wrong in a useful direction. The six-inch answer was absurd, and absurd is informative: it told the team which way to go before anyone bought pipe.
- Do the math on paper first. Changing a number on a page costs nothing; changing a pipe costs a trip to the store and a cut you can't undo.

DECISIONS MADE

- The profiler gets a bill of materials before anything is ordered. The bulk is on the page, and the known gaps stay visible until they are filled.
- A six-inch pipe is out. The first buoyancy pass showed it would take far too much ballast to bring a float that size to neutral, and the sizing moved to roughly a 4.5-inch outside diameter.
- Orders follow the list. Every sub-team's job this week is the same: finish the bill of materials, then get the parts ordered.

ACTION ITEMS

Finish the profiler bill of materials
Every sub-team · by 2.3

Order the parts once the bill of materials is closed
Management · by 2.3

Decide how the profiler talks to the topside (carried from 2.1)
Software group · by 2.3

STUDENT LEARNING OUTCOMES

- Built the first bill of materials for the vertical profiling float.
- Ran a first-pass buoyancy sizing calculation and used it to rule out an oversized hull.
- Applied the neutral-buoyancy relationship — weight equals displaced water — to a real design decision.
- Continued the payload manipulator design.

LOOKING AHEAD

Finish the bill of materials and get the parts ordered. The sizing calculation gets redone against real four-inch pipe dimensions, the profiler's communications decision is still waiting on the software track, and the manipulator design keeps moving.

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